

Serverless containers

Kubernetes-native serverless solutions allow developers to deploy containers without worrying about the underlying infrastructure. FOSS solutions like Knative, OpenFaaS, and Kubeless enable serverless workloads on Kubernetes.

Knative: A popular choice for serverless workloads, it auto-scales applications based on demand, including scaling down to zero during idle periods. It integrates natively with Kubernetes and provides event-driven capabilities.

OpenFaaS: Focuses on simplicity and ease of use, allowing developers to deploy both microservices and functions in a serverless manner on Kubernetes.

Kubeless: Lightweight and tightly integrated with Kubernetes, it allows serverless function deployment with full control over Kubernetes resources.

These solutions provide a flexible, cost-efficient serverless architecture by utilising Kubernetes clusters, scaling applications automatically, and eliminating the need for manual infrastructure management.

Serverless platforms: AWS Fargate and Google Cloud Run

Although Kubernetes is popular for managing containers, serverless platforms like AWS Fargate and Google Cloud Run have simplified container deployment by abstracting the infrastructure layer.

AWS Fargate: Fargate works with Amazon ECS and EKS, allowing developers to run containers without managing the underlying servers. It automatically provisions and scales compute resources, making it easier to handle varying workloads without worrying about servers.

- No need to manage EC2 instances.
- Built-in scaling and billing based on resource usage (vCPU and memory).
- Simplifies operational complexity for large container deployments.

Google Cloud Run: This is a fully managed serverless platform for stateless containers, supporting autoscaling based on HTTP traffic. Cloud Run simplifies deployment by running any containerised application without managing the infrastructure.

- Integrates with Google Kubernetes Engine (GKE) for hybrid cloud setups.
- Offers a ‘pay-per-use’ model, ensuring cost-effectiveness for scaling applications based on demand.
- Developers focus purely on code, as Cloud Run handles the infrastructure behind the scenes.

Both platforms abstract server management, making container deployments easier, more efficient, and cost-effective.

How AWS Fargate and Google Cloud Run simplify serverless architecture

Serverless platforms like AWS Fargate and Google Cloud Run play a pivotal role in simplifying container deployment by abstracting much of the underlying infrastructure management, allowing developers to focus more on their applications than on the operational overhead. Here’s how they do so.

▪ Eliminating infrastructure management

AWS Fargate: Fargate allows developers to run containers without having to provision or manage the underlying servers. It abstracts the complexities of scaling, patching, securing, and maintaining the infrastructure. Users only need to define the container image and the required resources, and Fargate handles everything from launching to auto-scaling the application.

Google Cloud Run: Similar to Fargate, Cloud Run is a fully managed platform that automatically scales stateless containers based on HTTP requests. There’s no need to manage VMs or Kubernetes clusters,

as Cloud Run handles the computer infrastructure.

▪ Auto-scaling based on demand

AWS Fargate: Fargate automatically scales containers up and down based on workload demands. This ensures that users pay only for the compute resources when the containerised applications are actually running, leading to cost efficiency.

Google Cloud Run: Cloud Run dynamically scales the container instances to handle requests, scaling down to zero when no requests are made, ensuring that resources are utilised efficiently without manual intervention.

▪ Flexible deployment options

AWS Fargate: Fargate can be integrated with both Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS), allowing users the flexibility to deploy containers using the orchestration platform of their choice without worrying about the infrastructure.

Google Cloud Run: Cloud Run supports deploying containers directly from Docker images, making it a highly accessible platform for developers who may not want the complexity of Kubernetes but still require fast, scalable container deployment.

▪ Pay-as-you-go model

AWS Fargate: Users are billed based on the resources consumed by the containers, such as CPU and memory usage, during the time the containers are running. There’s no need to over-provision resources in anticipation of traffic spikes.

Google Cloud Run: Google Cloud Run’s pricing model is also usage-based, billing only for the time containers are actively processing requests. This aligns cost with actual usage, making it economical for sporadic or variable workloads.

▪ Security and compliance

AWS Fargate: Fargate offers built-in security by isolating applications by design, running each task in its own microVM, ensuring a strong security